

Quiz: 02

Date: 28th December, 2025

Syllabus: Contrapositive, Proof by contradiction,

Proof by contradiction:

Theorem, $\boxed{P \Rightarrow Q}$
True,

'if, Q: is false, only way P: False

$P \Rightarrow Q$

Q: true or $\neg Q$: true

$(P \Rightarrow Q)$ True

$P \Rightarrow T$

$Q \Rightarrow T$, $P \wedge Q \Rightarrow T$

$P \Rightarrow T$

$Q \Rightarrow F$, $\neg Q \Rightarrow T$, $P \wedge \neg Q \Rightarrow T$

$P \Rightarrow Q$ (True)

P: True Q: False

Not possible,

$\neg Q \Rightarrow T \Rightarrow \Rightarrow \Rightarrow \dots \neg P \Rightarrow \text{True}$
but we said, P: True

• There does not exist a smallest positive rational number.

$$r \in \mathbb{Q} \quad r = \frac{p}{q} \quad , \quad p \in \mathbb{Z}, \quad q \in \mathbb{Z}, \quad q \neq 0$$

p & q are coprime

For the case of contradiction there exists

a smallest rational number, $r = \frac{a}{b}$ [$a: \text{int}, b: \text{int}$]

$a > 0$,

$$a > 0$$

$$\rightarrow a + a > a$$

$$b \neq 0$$

$$a \& b \text{ coprimes}$$

$$\rightarrow 2a > a$$

$$\rightarrow \frac{2a}{b} > \frac{a}{b}$$

$$b > 0, \quad b: \text{int}$$

$$2 > 0, \quad 2: \text{int}$$

$$\rightarrow \frac{a}{b} > \frac{a}{2b}$$

$$\rightarrow r > \frac{a}{2b}$$

a & $2b$ coprime

a & $2b$ not coprime $\frac{a}{2} = c$

$$r > \frac{c}{b}$$

$\therefore \frac{c}{b}$ is a rational number, r'

$\therefore r > r' > 0$ [we can form the same arguments if $a < 0$]

since, $a, b, 2$ all are positive integers

Thus we get a contradiction,

structure:

$$P \neq Q$$

Proof: Assume, $P: \top$, and $Q: \text{False}$, $\forall Q: \text{True}$

$\forall Q$ - - - - - $\rightarrow \neg P: \text{True}$
logic, Algebra

but we said,

$P: \text{True}$

no contradiction

Prove that,

if $A \in B$ are sets, $A \cap (B \setminus A) = \emptyset$

P

Q

$\neg Q: A \cap (B \setminus A) \neq \emptyset$

$\neg P: \text{either } A \text{ or } B \text{ or both are not sets,}$

Contrapositive statements very difficult to show!

We assume,

$$A \cap (B \setminus A) \neq \emptyset$$

so there is an element,

$$x \in A \cap (B \setminus A)$$

$$\Rightarrow x \in A \text{ and } x \in (B \setminus A)$$

$$\Rightarrow \underline{x \in A} \text{ and } (x \in B \text{ and } \underline{x \notin A})$$

so, contradiction,

$$\therefore A \cap (B \setminus A) = \emptyset$$

Proposition: Prove that,

there does not exist integer $m \in n$ for which

$$15m + 35n = 1$$

Proof: There does exist integers $m \in n$

for which $15m + 35n = 1$

$$\Rightarrow 5(3m + 7n) = 1$$

since, $m, n \in \mathbb{Z}$, $3m + 7n \in \mathbb{Z}$

$$3m + 7n = k \quad k \in \mathbb{Z}$$

$$\rightarrow 5 \cdot k = 1$$

$$k = \frac{1}{5}$$

so, contradiction

(Theorem: There are 'infinitely many primes')

$$p_1, p_2, \dots, p_k$$

$$n,$$

$$N = p_1 \cdot p_2 \cdot p_3 \cdot \dots \cdot p_n + 1$$

$$N = p_1 \cdot 2 \cdot \dots \cdot p_m + 1$$

use divisor algorithm,

Theorem: $\sqrt{2}$ is an irrational number,

sup, $\sqrt{2}$ is rational,

$$\rightarrow \sqrt{2} = \frac{p}{q} \quad [p, q \in \mathbb{Z}, q \neq 0, p \notin q \cdot \text{not coprimes}]$$

$$\rightarrow 2 = \frac{p^2}{q^2}$$

$$\rightarrow p^2 = 2q^2$$

p^r is even, p is even,

$$p = 2k \quad [k \in \mathbb{Z}]$$

$$\rightarrow (2k)^r = 2q^r$$

$$\rightarrow 2q^r = 2k^r$$

$$\rightarrow q^r = k^r$$

q^r is even, q is even

$\therefore p$ and q are coprimes

$\therefore \sqrt{2}$ is irrational

• $a \in \mathbb{Z}$, if a^r is even, then a is even,

Proof:

direct
proof
sketch

$$a \cdot b = \text{even}, \quad a = \text{even} \text{ or } b = \text{even}$$

$$b = a$$

$$a \cdot a = a^r = \text{even}, \quad a = \text{even} \text{ or } a = \text{even}$$

let, $p: a^r$ is even (True)

& $Q: a$ is even (False)

$\neg Q: a$ is odd (True)

$$a = 2k+1 \quad [k: \text{integer}]$$

$$a^r = (2k+1)^r = 4k^r + 4k^{r-1}$$

$$a^r = 2(2k+2) + 1$$

$$a^r = 2m+1 \quad [m \in \mathbb{Z}]$$

Proposition: if $a^r - 2a + 7$ is even, then a is odd

a is even

$$a = 2k \quad [k \in \mathbb{Z}]$$

$$(2k)^r - 2 \cdot 2k + 6 + 1 = 2(2k^r - 2k + 3) + 1$$

Defn:

20) We say that a point $P = (x, y) \in \mathbb{R}^r$ is rational if both x and y are rational

$$P: (x, y) \in \mathbb{Q}^r$$

An equation $F(x, y) = 0$ is said to have a rational point, if $F(x_0, y_0) = 0$ where $x_0, y_0 \in \mathbb{Q}$

example:

$$f(x, y) = 0$$

$$x^2 + y^2 - 1 = 0$$

$$x=0, y=1, x=1, y=0, x=\frac{3}{5}, y=\frac{4}{5}$$

Now, show that,

The curve, $x^r + y^r - 3 = 0$ has no rational point

Proof: $x^r + y^r - 3 = 0$ has at least one rational point,
let it be (x_0, y_0)

$$x_0 = \frac{p_1}{q_1} \quad y_0 = \frac{p_2}{q_2} \quad [p_1, q_1, p_2, q_2 \in \mathbb{Z}]$$
$$[q_1, q_2 \neq 0, p_1, q_1$$

$$\left(\frac{p_1}{q_1}\right)^r + \left(\frac{p_2}{q_2}\right)^r - 3 = 0$$

and p_2, q_2 are
coprimes]

$$\rightarrow p_1^r q_2^r + p_2^r q_1^r = 3 q_1^r q_2^r$$

$$\rightarrow p_1^r q_2^r = 3 q_1^r q_2^r - p_2^r q_1^r$$

$$\rightarrow p_1^r = \frac{q_1^r}{q_2^r} (3 q_2^r - p_2^r)$$

$$\text{let, } \gcd(q_1, q_2) = k$$

$$q_1 = k m_1 \quad \underline{q_2 = k m_2}$$

$$[m_1, m_2 \in \mathbb{Z}]$$

$$\frac{q_1}{q_2} = \frac{m_1}{m_2}$$

$$p_1^r = \frac{m_1^r}{m_2^r} (3 k^r m_2^r - p_2^r)$$

$$P_1^r = m_1^r (3k^r - \frac{P_2^r}{m_2^r})$$

$$\gcd(P_2, q_2) = 1, \quad \gcd(P_2, km_2) = 1$$

$$\gcd(P_2, k) = 1$$

$$\gcd(P_2, m_2) = 1$$

so, contradiction,

$$P_1 \in \mathbb{Z}, \quad P_1^r \in \mathbb{Z}$$

$$\text{but } m_1^r (3k^r - \frac{P_2^r}{m_2^r}) \notin \mathbb{Z}$$

2nd method:

$$\frac{P_1^r}{q_1^r} + \frac{P_2^r}{q_2^r} = 3$$

$$\rightarrow P_1^r q_2^r + P_2^r q_1^r = 3 q_1^r q_2^r$$

$$\text{let, } \gcd(q_1, q_2) = k \quad \left| \quad \text{let}$$

$$k \mid q_1 \quad \text{and} \quad k \mid q_2$$

$$q_1 = km_1, \quad q_2 = km_2, \quad \gcd(m_1, m_2) = 1$$

$$\rightarrow p_1^r k^r m_2^r + p_2^r k^r m_1^r = 3 k^r m_1^r m_2^r$$

$$\rightarrow p_1^r m_2^r + p_2^r m_1^r = 3 m_1^r m_2^r$$

$$\rightarrow \underbrace{(p_1 m_2)^r}_a + \underbrace{(p_2 m_1)^r}_b = 3 \underbrace{(m_1 m_2)^r}_c$$

$a, b, c \in \mathbb{Z}$ Why!

$\gcd(a, b, c) = 1$ how!

$$c = m_1 \cdot m_2 \quad b = p_2 m_1 \quad , \quad a = p_1 m_2$$

$$\gcd(b, c) = m_1$$

$$\gcd(a, \gcd(b, c)) = \gcd(p_1 m_2, m_1) = 1$$

$$a^r + b^r = 3c^r$$

$$a^r + b^r \equiv 0 \pmod{3}$$

$$a^r \equiv 0 \pmod{3} \quad b^r \equiv 0 \pmod{3}$$

$$3 \mid a^r \quad 3 \mid b^r$$

prime $\mid \alpha\beta$, prime $\mid \alpha$ or prime $\mid \beta$, or both

$$3 \mid a$$

$$3 \mid b$$

$$a = 3k_1$$

$$b = 3k_2$$